Same Model, New Patient: Can Synthetic Heartbeats Rescue Rare-Arrhythmia Detection?

Rare-Arrhythmia Detection with Generative ECG Augmentation

Course project, short paper

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Abstract

Tested on beats of the patients it was trained on, our heartbeat classifier detects the rare arrhythmia classes with a rare-class F1 of 0.837. Tested on patients it has never seen, it scores 0.088. We asked whether class-conditional generative models, a conditional VAE and a WGAN-GP, can close that gap by synthesising the rare beats that are missing. Under rules registered before any test record was read, with training and test patients strictly apart, we compared both generators with no augmentation, a class-weighted loss, random oversampling and SMOTE for two classifiers, and read the 22 test patients once. Generative augmentation did not detectably help: the best case, the CNN with the cVAE, gained +0.025 (95% CI −0.018 to +0.047). Twenty-three pre-registered follow-up analyses then found what does help: describing each beat relative to the patient's own beats, judging prematurity against the patient's own rhythm, vetoing S calls during atrial fibrillation and averaging ten runs raised the test score to 0.587, level with the best published inter-patient result. Every prediction comes with a saliency map and a note whose every claim is checked against the model's output.

1 Introduction

A heartbeat classifier can look excellent or useless on the same data. Trained on the MIT-BIH Arrhythmia Database [11] with its beats split at random, our convolutional network finds the rare arrhythmia beats with a rare-class F1 of 0.837. Trained on one group of patients and tested on another, the same network scores 0.088. Nothing about the network changed; only the patients did (Figure 1).

The reason is that patients differ. A heartbeat's shape depends on the heart, the electrode position and the disease, and the beats that matter most are both rare and concentrated in a few people: supraventricular ectopic beats (S), whose burden goes with later atrial fibrillation and stroke [2], and fusion beats (F). A model that has seen a few hundred S beats from a dozen people has to recognise them in someone new.

Generative models offer an apparent way out: learn what the rare beats look like and manufacture more of them. Conditional VAEs [13] and Wasserstein GANs [6] can draw convincing synthetic ECG, and augmenting with them is often reported to help, mostly on splits where the test patients were also in training. We asked the question a clinic would ask:

> *With training and test patients strictly apart, does class-conditional generative augmentation improve detection of the rare classes S and F, compared with no augmentation, a class-weighted loss, random oversampling and SMOTE?*

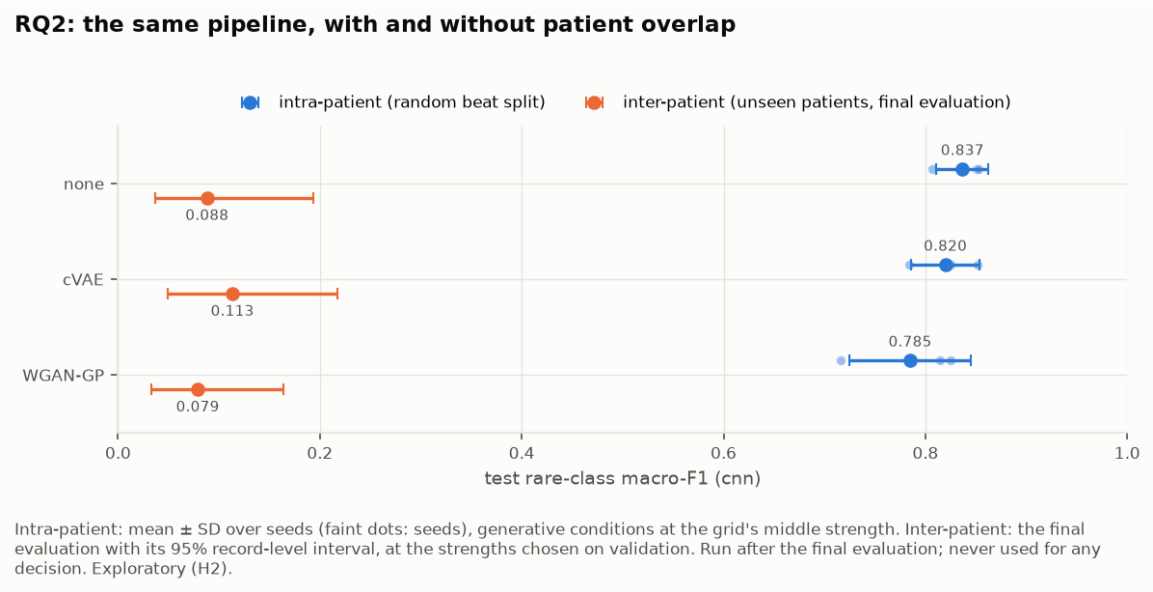


Figure 1: The same pipeline, scored on held-out beats of the training patients (blue) and on patients it has never seen (orange). The rare-class macro-F1 falls from about 0.8 to about 0.1, with or without generative augmentation.

We built the whole pipeline (data, two classifiers, two generator families, augmentation, saliency maps and fact-checked notes) and ran it in phases. Each phase ended with a gate at which a rule registered in advance chose on validation patients only, and the test patients were read once, at the end; 761 runs are logged. This paper reports four findings:

- **The pre-registered answer is no.** Generative augmentation did not detectably improve rare-class detection on new patients; it performed on par with the cheap balancing methods (Section 4.1).
- **Patient overlap inflates the score almost tenfold**, so scores from random beat splits say little about new patients (Section 4.2).
- **Realistic synthetic beats are not useful beats**: no synthetic-quality metric predicted the gain (Section 4.3).
- **What does help is knowing the patient**: 23 follow-up analyses, each registered before it ran, raised the test score from 0.113 to 0.587 and tested it on 977 patients no analysis had used (Section 4.4).

2 Data and rules

Data. The MIT-BIH Arrhythmia Database [11, 5] holds 48 half-hour two-lead recordings whose beats were labelled by cardiologists. We use lead MLII at 360 Hz, map the labels to the AAMI classes [1] N (normal), S (supraventricular ectopic), V (ventricular ectopic) and F (fusion), and leave out the four paced records. Figure 2 shows what the classes look like.

Patients kept apart. We use the split of de Chazal et al. [4]: 22 records (DS1) for training and validation, the other 22 (DS2) for the test, so no patient is in two splits; five DS1 patients form the

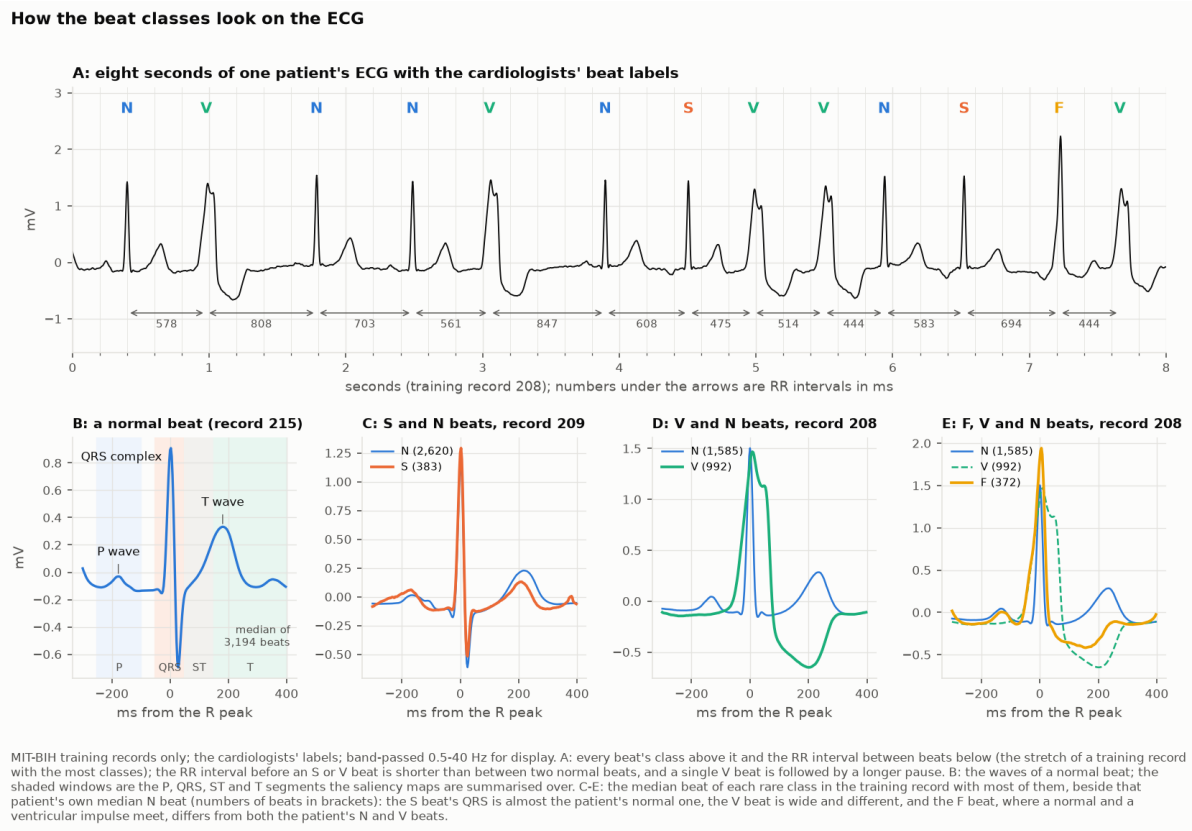


Figure 2: The beat classes on the ECG (training records only). An S beat comes early but looks almost like the patient's normal beat; a V beat is wide and different; an F beat lies between the two. The RR intervals (arrows) are what reveals an early beat.

validation set (Table 1). Generators, oversamplers and normalisation see training patients only.

Split	Records	N	S	V	F
training	17	35,496	644	3,030	388
validation	5	10,370	300	758	27
test (DS2)	22	44,259	1,837	3,221	388

Table 1: Records and beats per split. S beats are under 4% of each split, and F beats at most 1%.

Measure and rules. The primary measure is the rare-class macro-F1, the mean F1 of S and F on the test patients, with 95% confidence intervals from a bootstrap over patients and Holm's correction [7] for the four confirmatory comparisons. Every selection rule and hypothesis was registered before the experiment it governs, and each phase ended with a gate at which the rule was applied and approved. The test file opens only for the final evaluation, and every read is logged. The follow-up analyses were registered as new versions of the rules (21 in all) before they ran, and are reported apart from the primary results.

3 The system

Figure 3 shows the pipeline. Each stage was chosen on validation patients:

- **Beat:** a 0.5 s window around the R peak, z-scored, with the RR intervals before and after the beat as extra input (the best of four representations).
- **Classifiers:** a shallow CNN (73k parameters) and a BiLSTM (135k), the best of four architectures; pretraining on PTB-XL did not help.
- **Generators:** a conditional VAE and a WGAN-GP, each the best of four designs, trained with three seeds into frozen banks of synthetic beats.
- **Augmentation:** each rare class brought up to a fraction (0.1, 0.25 or 0.5) of the N count with synthetic beats, keeping those closest to real training beats by dynamic time warping; the fraction chosen per condition. With no augmentation, the class-weighted loss, random oversampling and SMOTE [3] this made a 96-run grid.
- **Explanation and note:** Grad-CAM [12] for the CNN and occlusion for the BiLSTM, summarised as the share of attention in the P, QRS, ST and T windows; a language model (Gemini) writes a note from a JSON of the model's output only, and a fact checker rejects any number, class or finding not in that JSON.

4 Results

Four hypotheses were registered before any test record was read (Table 2); the verdicts are the registered rules applied as written. The sections below give the evidence behind each.

Hypothesis	Statement	Verdict	Evidence
H1	confirmatory: a generative condition improves test rare-class macro-F1 over none (Holm, 4 tests)	not supported	no confirmatory comparison significant
H1b	exploratory: gains larger for F than for S	mixed	F gain larger than S gain in 2 of 4 generative conditions
H1c	secondary: generative beats SMOTE, random oversampling and class-weighted loss	not supported	0 of 6 comparisons of the best generative condition with a baseline significant
H2	exploratory: intra-patient evaluation inflates the baseline and the gains	not consistent	inflation of the baseline +0.748; larger intra-patient gain for 0 of 2 generators
H3	exploratory: a quality metric predicts the downstream gain	not supported	0 of 5 metrics with an interval excluding zero
H4	exploratory: augmented models are at least as faithful as the baseline	mixed	augmented faithfulness at least the baseline's in 1 of 4 model pairs

Table 2: The pre-registered hypotheses, their verdicts and the evidence, all on the 22 test patients.

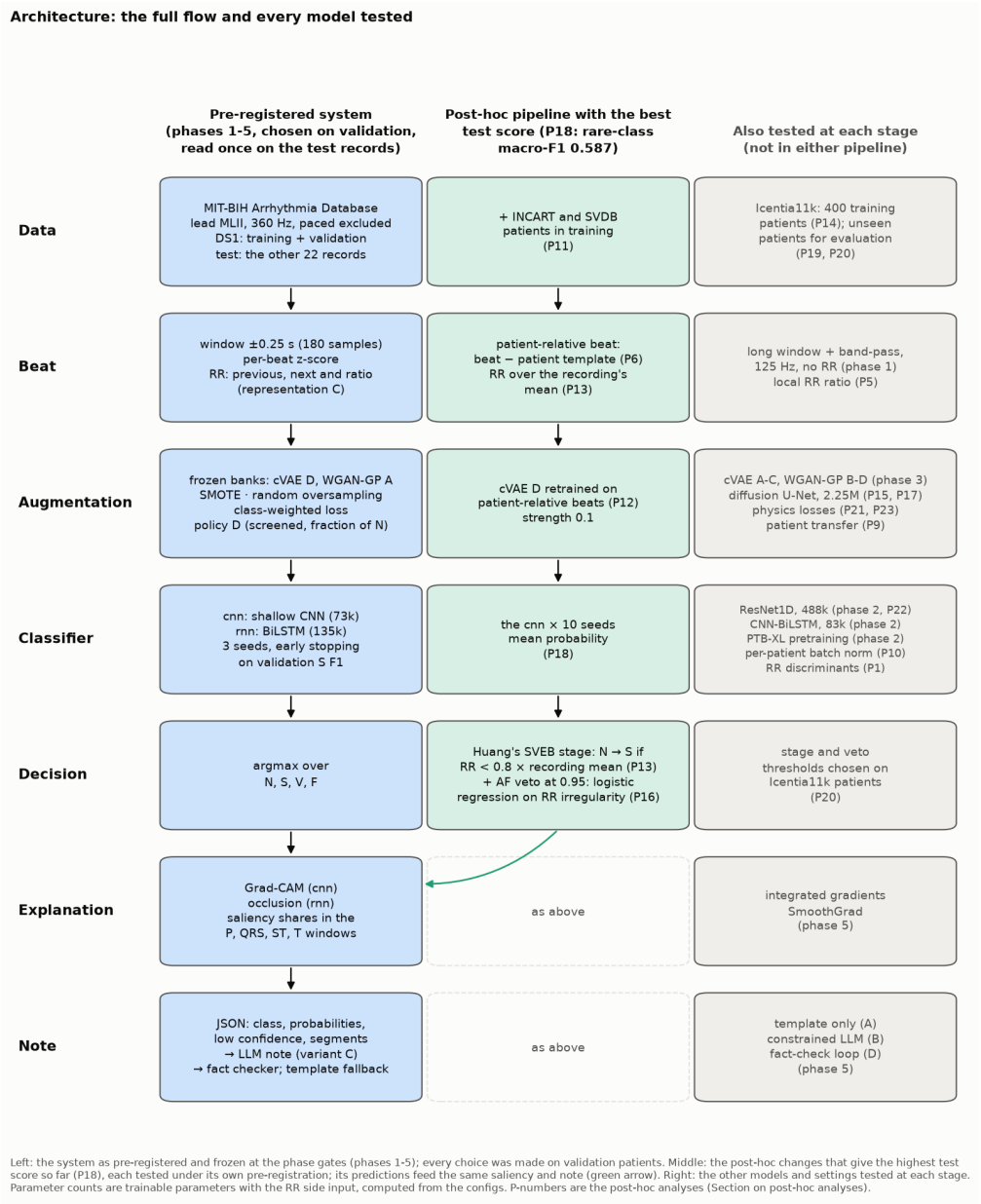


Figure 3: The pipeline. Left: the pre-registered system. Middle: the follow-up pipeline with the best test score (Section 4.4). Right: what else was tested at each stage.

4.1 The pre-registered answer

Table 3 gives the single test read. For the CNN, training without augmentation scored 0.088, the cVAE 0.113 and the WGAN-GP 0.079, against 0.113 for the class-weighted loss and 0.136 for random oversampling. The cVAE's gain over no augmentation, +0.025 (95% CI −0.018 to +0.047), points the right way, and it nearly tripled the CNN's sensitivity for F beats, from 0.064 to 0.185. But no confirmatory comparison survived Holm's correction, and the cVAE was level with the cheap methods (+0.001 against the class-weighted loss, −0.022 against random oversampling). For the BiLSTM the differences were smaller still. The hypothesis is not supported at this sample size.

Architecture	Condition	Strength	Rare macro-F1 [95% CI]	N recall
cnn	none	–	0.088 [0.037, 0.194]	0.962
cnn	class weighted	0.5	0.113 [0.057, 0.207]	0.933
cnn	random oversampling	0.1	0.136 [0.061, 0.247]	0.844
cnn	smote	0.1	0.070 [0.025, 0.172]	0.985
cnn	cvae	0.25	0.113 [0.049, 0.217]	0.938
cnn	wgan gp	0.1	0.079 [0.033, 0.164]	0.955
rnn	none	–	0.075 [0.014, 0.232]	0.961
rnn	class weighted	0.75	0.061 [0.022, 0.168]	0.865
rnn	random oversampling	0.1	0.090 [0.028, 0.196]	0.883
rnn	smote	0.1	0.086 [0.025, 0.206]	0.894
rnn	cvae	0.1	0.073 [0.035, 0.157]	0.926
rnn	wgan gp	0.5	0.069 [0.027, 0.156]	0.857

Table 3: The 22 test patients, read once: rare-class macro-F1 with 95% intervals from a bootstrap over patients, and the recall of normal beats. The strength is the fraction of the N count chosen on validation.

4.2 Patient overlap inflates the score

After the final evaluation, the same CNN was trained and tested on a random beat split of all 44 records. Without augmentation it reached 0.837 ± 0.026 (Figure 1), and 0.803 and 0.777 on two further random splits. The generators did not help there either (cVAE −0.017, WGAN-GP −0.052). A model tested on the patients it was trained on is largely tested on its memory of those patients: published results with random splits above 0.9 [9] and inter-patient results around 0.5–0.6 [8, 15] measure different things.

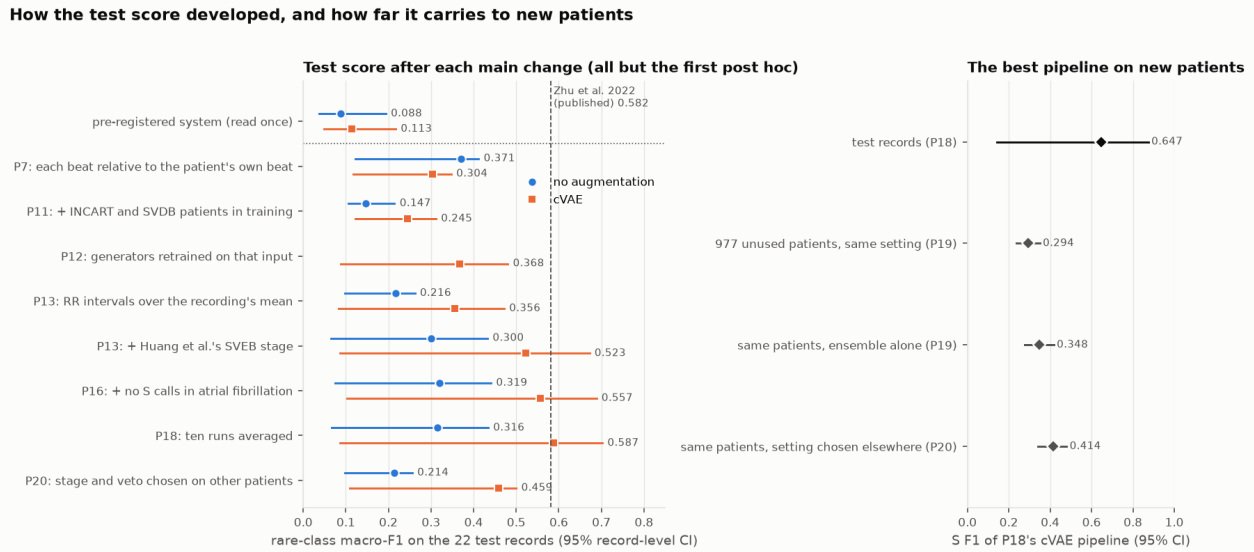
4.3 Realistic synthetic beats are not useful beats

We first checked that the quality metrics rank deliberately degraded beats correctly (dynamic time warping and a spectral distance did; a Fréchet-distance analogue and train-on-synthetic-test-on-real did not). Five metrics, covering fidelity, diversity, memorization and a classifier two-sample test, were then compared with the downstream gain over 96 checkpoints of 24 generator runs. None predicted it (Spearman ρ from −0.20 to +0.18, every interval including zero). A follow-up analysis shows why: 57–65% of the synthetic S beats lie nearest to a single training patient, record 209, just as 60% of the real training S beats do, while the validation patients' S beats lie nearest to training patients who contribute only 1–2% of them. The generators reproduce the training patients faithfully, and that is not what new patients need.

4.4 What does help: knowing the patient

After the final evaluation, 23 follow-up analyses (P1–P23), each registered before it ran and reported apart, asked what does move the score. Four changes mattered (Figure 4):

1. **Each beat relative to the patient** (P7): subtracting the patient's own template beat raised the CNN without augmentation from 0.088 to 0.371.
2. **Prematurity against the patient's rhythm** (P13): RR intervals divided by the recording's mean RR [10], and Huang et al.'s rule that calls an early beat S [8], raised the cVAE condition to 0.523.
3. **No S calls during atrial fibrillation** (P16): a detector using RR intervals only, fitted on the MIT-BIH Atrial Fibrillation Database [5], raised every condition by 0.018 to 0.034, each 95% interval above zero.
4. **Ten runs averaged** (P18): the cVAE's ten-run ensemble reached 0.587 [0.086, 0.702], with S F1 0.647 and F F1 0.527.



Left: the cnn without augmentation (circles) and with the cVAE (squares); the first row is the pre-registered system, every row below the dotted line a post-hoc analysis registered before it ran. P12 retrains only the generators, so it has no row without augmentation. All rows are read on the same 22 test records, so the later rows are optimistic. Dashed: the mean of S and F F1 reported by Zhu et al. (2022). Right: S F1 of P18's best pipeline (the cVAE's ten-run ensemble with the SVEB stage and the AF veto) on the test records (record-level CI) and on Icentia11k patients that no analysis had used (patient-level CI); those patients have no F labels.

Figure 4: Left: the test score after each main change, without augmentation (circles) and with the cVAE (squares); every row below the dotted line is a follow-up analysis. Right: the best pipeline's S F1 on the test patients and on unseen Icentia11k patients.

That places the best pipeline level with the best published inter-patient result (Table 4). It is also optimistic: the follow-up analyses read the same 22 test patients many times. On 977 Icentia11k patients [14] that no analysis had used, the same pipeline reached S F1 0.294 [0.237, 0.350], and 0.414 when its two thresholds were chosen on 388 other Icentia11k patients; on the test records that choice gave 0.459, because one test record (232) holds 1,381 of the 1,836 usable test S beats and no tuning patient resembles it. The cVAE gave the highest test score in P13, P16 and P18, but on the unseen patients it was level with the other conditions (differences from −0.001 to +0.008, none detectable).

Inter-patient (DS2) results	S F1	V F1	F F1	Mean of S and F F1
de Chazal et al. 2004 [4]	0.511	0.797	–	–
Huang et al. 2014 [8]	0.577	0.924	–	–
Zhu et al. 2022 [15]	**0.686**	**0.960**	0.477	0.582
Ours, pre-registered (CNN with the cVAE)	0.173	0.893	0.054	0.113
Ours, follow-up P18 (cVAE ensemble, S stage, AF veto)	0.647	0.936	**0.527**	**0.587**

Table 4: Beside published inter-patient results (F1 computed from the reported sensitivity and positive predictivity where a paper gives no F1).

4.5 A prediction a clinician can check

Of four saliency methods, the rule chose Grad-CAM for the CNN (the most faithful: deleting its most salient samples hurt the model most) and occlusion for the BiLSTM. Of four note generators tried on 50 beats, all made zero unsupported claims; the rule chose Gemini with worked examples and a fixed output schema, which covered every required fact in 2.6 s per note. The demo (Figure 5) shows the whole chain for one beat: the offline analysis by our models, then a review written by Gemini from those results only, with every claim it makes checked against them.

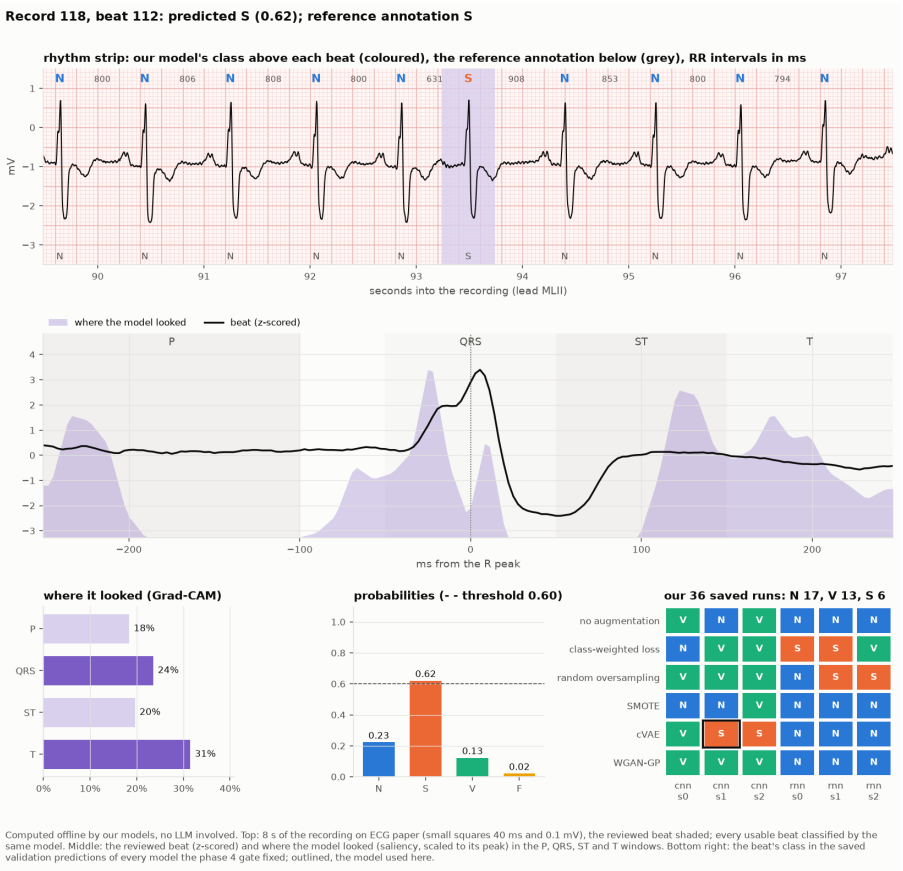


Figure 5: The demo for one S beat of a validation patient: the rhythm strip with each beat's predicted class and the reference label, where the model looked, the probabilities, and how all 36 saved runs classified this beat (only 6 called it S).

5 What it means

Variety, not volume. With 17 training patients, the bottleneck for new patients is not the number of rare beats but how many different patients they come from. The generators multiplied the beats of the few patients they saw; what helped instead put each beat in the context of its own patient, as a cardiologist reads a rhythm strip: is this beat early for this heart, does it differ from this patient's normal beat, is the rhythm fibrillating?

The protocol decides the score. The same network scores 0.837 or 0.088 depending only on whether the test patients were seen in training. Results on random beat splits should not be compared with results on new patients.

Limitations. The rare beats sit in very few patients (most test S beats in one record), so the intervals are wide; records 201 and 202 come from the same person (the conclusions hold without record 202). The follow-up analyses are optimistic because they read the same test patients; the unseen Icentia11k patients give the fairer estimate. The notes restate the model; they do not diagnose.

Conclusion. Under a strict inter-patient protocol, synthetic heartbeats from a cVAE or a WGAN-GP did not detectably improve rare-arrhythmia detection over simple balancing. Patient-relative inputs, rhythm context and ensembles did, reaching the level of the best published inter-patient result. Showing whether generative augmentation adds to that needs what it cannot manufacture: more patients with rare beats.

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